Disagreement in r^*?

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1 The question

Long-term bond yields are anchored by beliefs about where short rates settle in the long run. A revision to that endpoint passes into the yield at every maturity, undamped by horizon, in a way that news about the current policy stance does not. This is why the endpoint has been at the center of the term-structure literature since Kozicki and Tinsley's shifting endpoints, and why Bauer and Rudebusch's falling stars put a slow-moving anchor at the heart of it. What is unsettled is where those beliefs come from, and whether the central bank is one of the sources. Consider a policy rule written as follows,

$$i_t \; = \; \underbrace{r_t^* + \pi_t^*}_{\text{intercept}} + \phi_\pi(\pi_t - \pi_t^*) + \phi_x x_t + v_t, \tag{1}$$

r^* is the *intercept* of the reaction function, so "the Fed reveals r^*" means the Fed is telling markets about the constant in its own rule. Bauer, Pflueger & Sunderam (2024) estimate the perceived *slope* ϕ_π and show it prices term premia; they absorb the intercept into forecaster fixed effects. The intercept has been assumed known, assumed common, or differenced away.

> **How does the Fed move beliefs about the intercept of its own reaction function, and what does disagreement about that intercept do to the price of long bonds?**

In three parts:

1. **Who moves whom**: Does the FOMC's published longer-run projection move primary dealers' beliefs about the same object, and by how much? And does the dealers' projection, which the Desk reports to the Committee before every meeting, move the FOMC's own, and by how much?

2. **Pricing**: Does a revision to the published projection move the yield curve with the maturity profile of an endpoint shock? Can the term-premium response be separated from stance news and from a perceived policy error?

3. **Disagreement**: Does dispersion in beliefs about the intercept carry a risk premium whose sensitivity does not decay with maturity? Does a signed Fed-minus-market gap move the level of yields, with the split between breakeven and real-rate responses pinned by the perceived rule?

2 Decomposing Long Yields

Start from the definition of a long yield as an average of expected short rates plus a residual:

$$y_t^{(n)} = \underbrace{\frac{1}{n}\sum_{j=0}^{n-1} E_t[i_{t+j}]}_{\text{expectations component } rn_t^{(n)}} + TP_t^{(n)}. \tag{2}$$

Now split the nominal policy rate into a neutral level and a deviation from it. Define the *neutral nominal rate* and the *policy gap*

$$i_t^* \equiv r_t^* + \pi_t^*, \qquad \text{gap}_t \equiv i_t - i_t^* = \underbrace{(i_t - E_t\pi_{t+1} - r_t^*)}_{\text{real policy stance}} + \underbrace{(E_t\pi_{t+1} - \pi_t^*)}_{\text{inflation deviation}}, \tag{3}$$

where r_t^* is the equilibrium real rate and π_t^* the long-run inflation anchor. The gap is therefore **the stance of policy**: how far the Fed is holding the short rate above or below the level that would be neutral, plus how far inflation is expected to run from its anchor. Under a Taylor-type rule $i_t = i_t^* + \phi_\pi(\pi_t - \pi_t^*) + \phi_x x_t + v_t$, the gap is exactly the systematic and discretionary response to the cycle,

$$\text{gap}_t = \phi_\pi(\pi_t - \pi_t^*) + \phi_x x_t + v_t. \tag{4}$$

Substituting (3) into (2), and adding the martingale assumption for the endpoints — $E_t[r_{t+j}^*] = r_t^*$ and $E_t[\pi_{t+j}^*] = \pi_t^*$ — gives

$$y_t^{(n)} = r_t^* + \pi_t^* + \frac{1}{n}\sum_{j=0}^{n-1} E_t\big[\text{gap}_{t+j}\big] + TP_t^{(n)}. \tag{5}$$

Equivalently, in terms of the neutral nominal rate,

$$y_t^{(n)} = i_t^* + \frac{1}{n}\sum_{j=0}^{n-1} E_t\big[\text{gap}_{t+j}\big] + TP_t^{(n)}. \tag{6}$$

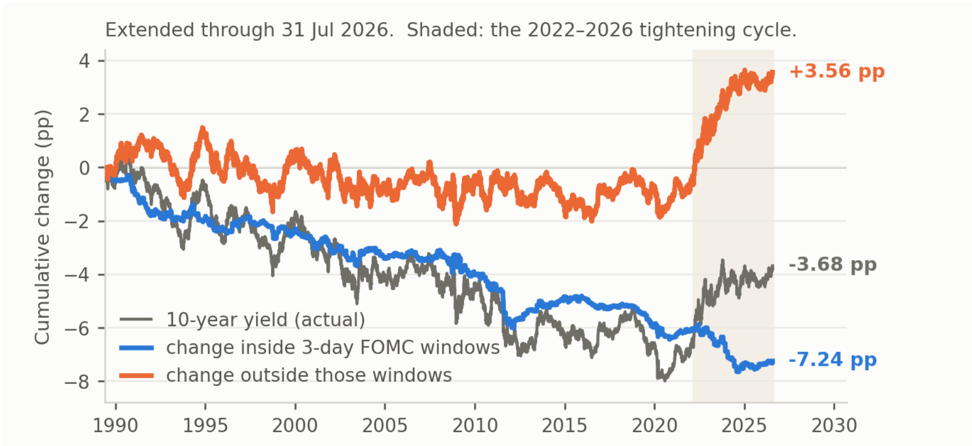

Figure 1: Fact 1. Cumulative change in the ten-year yield from 2 June 1989, split into the part accruing inside three-day FOMC windows and the part accruing on all other days. Shaded: the 2022–2026 tightening cycle.

3 Three facts

3.1 Fact 1: through 2021, yield changes outside FOMC windows net to zero

Replicating Hillenbrand (2025) on GSW yields and a hand-built event list: yields move on non-FOMC days about as much as on FOMC days — daily standard deviations of 5.60 against 6.72 bp. What differs is that outside the window the moves offset. Across 7,035 non-FOMC days the cumulative change is -0.82 pp, a drift of -0.012 bp/day with $t = -0.18$; inside three-day FOMC windows, 11.8% of trading days, the ten-year fell 6.16 pp of a total 6.99. Hillenbrand's explanation is **long-run Fed guidance**: announcements are the moments at which the Fed reveals its own estimate of where rates settle in the long run — implicitly before 2012, when the rate decision itself disclosed the Committee's read of the natural rate, and explicitly after 2012 through the SEP longer-run dot.

$$y_t^{(n)} \;=\; i_t^* \;+\; \frac{1}{n}\sum_{j=0}^{n-1} E_t\big[\text{gap}_{t+j}\big] \;+\; TP_t^{(n)},$$

Hillenbrand's reading is that window moves are Δi^*, which requires the other two terms to contribute little. For the stance he appeals to the far-forward evidence: the same downward trend appears in the 5-year, 5-year-forward rate, where the current stance should long since have washed out. For the term premium he reports in the Internet Appendix that model-based estimates *do* fall in these windows, and sets the finding aside on the grounds that those models cannot represent a secular decline. A third result, that the decline since 1999 is in TIPS rather than breakevens, does not exclude a term but tells us the composition of Δi^*: it is the real endpoint moving, not the inflation anchor.

3.2 Fact 2: the two beliefs move together, and asymmetrically

The FOMC publishes its longer-run funds rate projection quarterly; the NY Fed surveys primary dealers for the same object about two weeks before every meeting. Netting the 2% objective from both gives two comparable measures of r^* (Figure 2). Across 48 matched meetings (46–47 for the revision regressions): the gap has a standard deviation of 0.105 pp and a range of -0.11 to $+0.30$, exceeding 25 bp at 4% of meetings; dealers move with the Fed nearly one-for-one ($\beta = 0.82$, $t = 6.86$); the reverse relation is present but weaker ($\beta = 0.31$, $t = 3.91$). The asymmetry is not

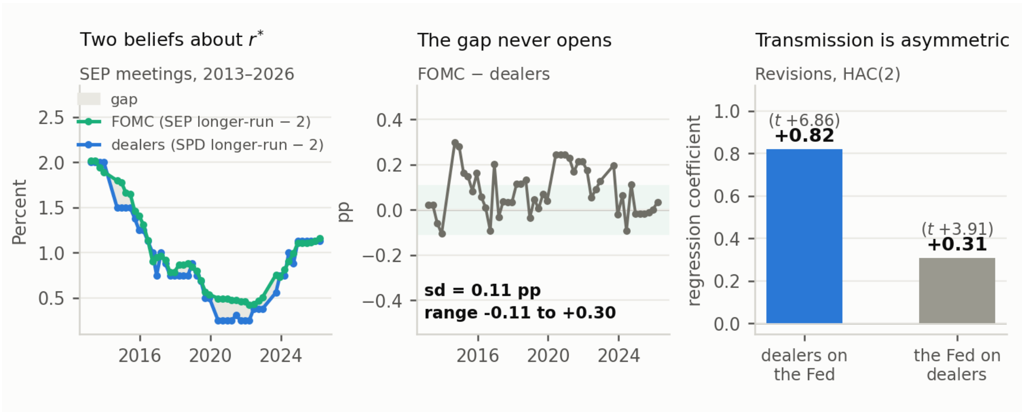

Figure 2: Fact 2. Left: the FOMC's longer-run projection and the primary dealers' longer-run funds rate, each less 2.0. Centre: the difference. Right: revision regressions, HAC(2), 46–47 meetings.

mechanical — the Desk briefs the FOMC with survey results before each meeting — but it is not a clean lead-lag either. Dealer revision loads on both the preceding dot revision (0.60) and the contemporaneous one (0.58), so part of what looks like following is anticipation; and intermeeting macro news moves both series, so these regressions describe timing, not identified transmission. Both series are observed eight times a year. Section **??** adds a pricing counterpart to this fact: the announcement window prices exactly the part of the dot revision that the dealers' own preceding revision did not anticipate.

3.3 Fact 3: the FOMC's r^* and the econometric r^* move in opposite directions

The FOMC's estimate can be compared against the leading model-based alternative. Holston, Laubach & Williams estimate the equilibrium real rate from a semi-structural macro model with a Kalman filter, quarterly from 1961. Under the martingale assumption, these two are estimates of the same object: HLW's contemporaneous r^*_t and the FOMC's longer-run projection coincide when $E_t[r^*_{t+j}] = r^*_t$. They do not behave like estimates of the same object (Figure 3). Across the 57 SEP meetings the correlation of levels is $-\mathbf{0.93}$. Between January 2012 and March 2022 the FOMC marked its own r^* *down* by 1.78 pp, from 2.21% to 0.42%, while HLW marked r^* *up* by 1.37 pp, from 0.89% to a peak of 2.26%. After 2022 they swap: the FOMC revises up 0.74 pp while HLW falls back. The two series cross exactly once, in June 2016, and the mean absolute gap between them is 0.88 pp — they sit within 25 bp of each other at only 5% of meetings. Three things follow. First, the Fed's r^* is not easily recoverable from an econometric model. Second, the same holds for the market side: it is hard to justify a model-based endpoint to serve as the market perception of r^*. Third, the premise that the Fed holds an information advantage about r^* has an observable consequence: over the decade in which the Committee cut its estimate by 1.78 pp, the filtered econometric estimate moved the other way. The comparison uses one-sided estimates under current parameters; whether it survives on the real-time vintages, which declined over 2012–2019, and on the post-COVID revision of the model is being verified. One candidate explanation can be rejected immediately. HLW estimates r^* from a model; the Fed's is inferred from its reported i^* and π^*, and the SEP reports $\pi^* = 2.0\%$ with zero variation over the observed sample. If participants report i^* sincerely but hold π^* at 2% to anchor expectations, the netting $r^*_{\text{Fed}} = \text{dot} - 2$ would mislabel the Fed's real endpoint and the two series would not be comparable. Figure 4 tests this directly: it computes the long-run inflation expectation that would have to be embedded in the FOMC's projection for the two estimates to agree. That implied series runs from 3.46% in September 2012

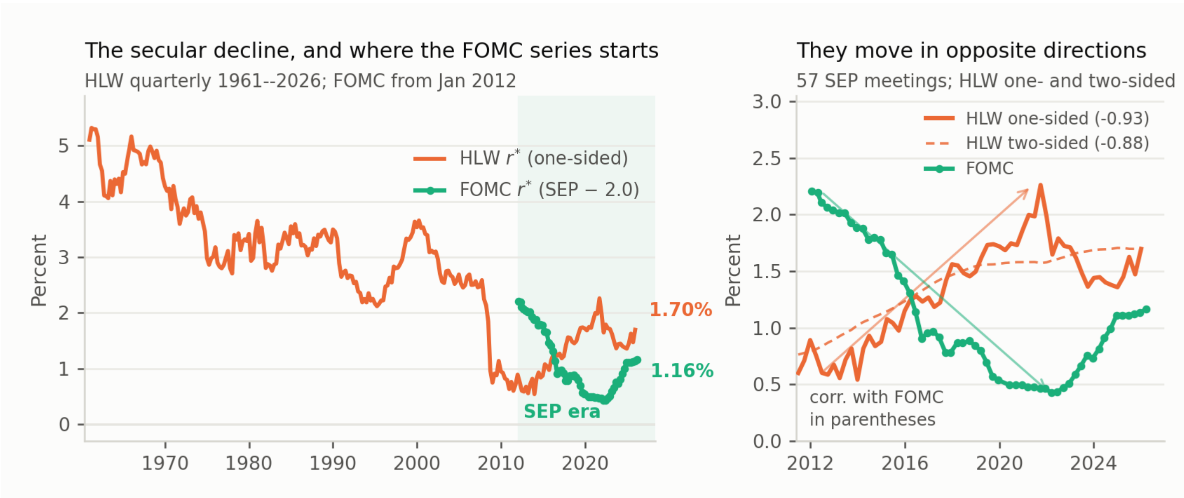

Figure 3: Fact 3. Left: HLW r^* over the full quarterly sample with the FOMC series, which begins in January 2012, overlaid; the SEP era is shaded. Right: the overlap sample, with arrows marking the direction each series travels over 2012–2022.

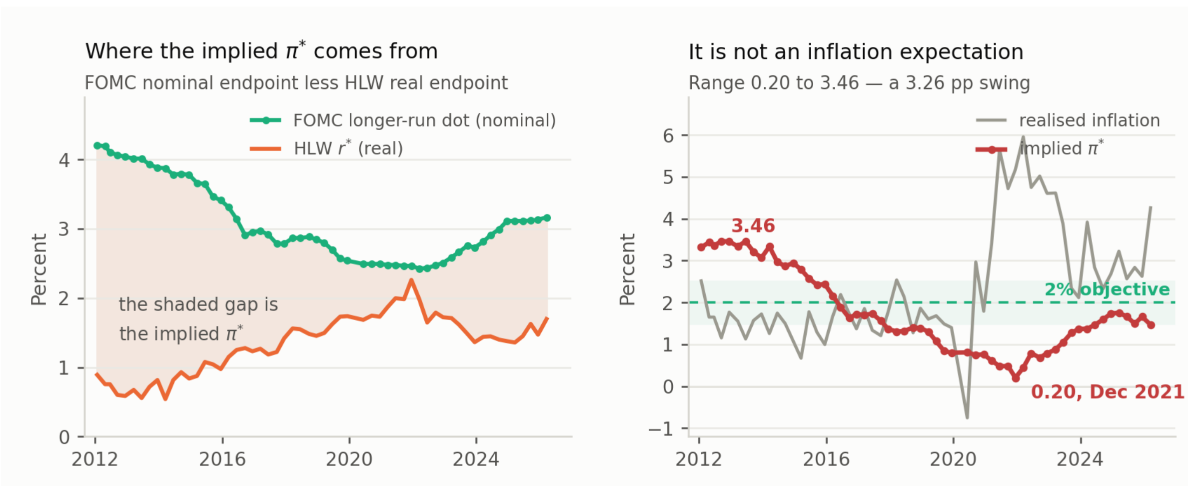

Figure 4: The inflation expectation needed to reconcile the two estimates. Left: the FOMC's longer-run federal funds projection, which is nominal, against HLW's one-sided r^*, which is real, at the 57 SEP meetings; the shaded area between them is the long-run inflation expectation that would have to be embedded in the FOMC's projection for the two series to be mutually consistent. Right: that implied series plotted against realised inflation, with the 2% objective marked and a 1.5–2.5% band shaded. The implied series lies outside the 1.5–2.5% band at 75% of meetings. Realised inflation is HLW's own four-quarter input series rather than headline PCE.

to 0.20% in December 2021 — at its trough it requires FOMC participants to have expected 0.20% long-run inflation at a moment when realised inflation was 5.17%. Whatever separates the two estimates, it is not the 2% netting.

4 Framework: intercept disagreement in an opinionated-markets model

Caballero & Simsek (2022) study a Fed and a market that disagree, each confident in its own view, with the Fed setting policy on its belief and the market pricing assets knowing it will. The market therefore anticipates policy "mistakes" and prices the resulting risk. Their state variable is aggregate demand. This section runs the same logic with the disagreement located in the *intercept of the policy rule.* What distinguishes intercept disagreement is not its persistence — the estimated Fed–dealer gap mean-reverts with a half-life of about six months — but that it maps into a *composition* restriction across the nominal and real curves that demand disagreement does not generate.

4.1 Setup

Both parties know the rule and observe the state. They differ only in the neutral real rate: the Fed believes r^{*F}, the market believes r^{*m}, and

$$\delta_t \;\equiv\; r_t^{*m} - r_t^{*F}. \tag{7}$$

The Fed sets policy on its own estimate,

$$i_t \;=\; r_t^{*F} + \pi^* + \phi_\pi(\pi_t - \pi^*) + \phi_x x_t, \qquad \phi_\pi > 1, \tag{8}$$

and the market evaluates the resulting stance against *its* neutral rate, perceiving policy to be δ_t looser than the Fed intends. Three roles separate whose belief enters where:

	Role	**Estimate used**
Fed	Sets i_t through the rule (8)	r^{*F}
Economy	Adjusts π until the real rate reaches the natural rate	the truth, r^*
Market	Forecasts the path of i; prices bonds off that forecast	r^{*m}

The market's error is that it substitutes its own r^{*m} for the second row when it runs the economy forward in its head.

4.2 The market's subjective steady state

Consider the steady state the market *forecasts*, holding its belief dogmatically. Two conditions define it. Over the long run the Fed closes the output gap, so $\phi_x x_t$ drops out and permanent accommodation is absorbed entirely by inflation rather than output; and the natural rate is by definition the real rate consistent with a zero gap, so $i - \pi$ equals the natural rate — which, in the market's forecast, is the one the market believes in. Setting $i - \pi = r^{*m}$ in (8) gives $(\phi_\pi - 1)(\pi - \pi^*) = \delta$, so the market expects inflation to settle permanently away from target at

$$\boxed{\pi - \pi^* \;=\; \frac{\delta_t}{\phi_\pi - 1}} \tag{9}$$

The target is a parameter of the rule, held at 2% throughout the sample — all 594 archived longer-run PCE submissions are exactly 2.0, with zero within-meeting dispersion. What moves is *realised* long-run inflation. The Fed does not revise its target; it persistently misses one it continues to hold, because it is setting policy against the wrong intercept. The adjustment runs through the real rate: at $\pi = \pi^*$ the real rate is r^{*F}, below the natural rate, so demand exceeds supply and inflation rises; each point of inflation above target raises the real rate by $\phi_\pi - 1$, until the real rate meets the natural rate. Nobody chooses the natural rate and nobody chooses π — inflation is the free variable, and it converges to a finite bias only because $\phi_\pi > 1$. The beliefs are subjective, not actual: in the true steady state $i - \pi = r^*$, and (9) with δ replaced by $r^* - r^{*F}$ gives the realised bias, which coincides with the market's version only if the market happens to be right. Nothing below requires that it is — bonds are priced off $E_t^m[i_{t+j}]$, so (9) and (10) hold whether or not those expectations are correct. The macro logic is Orphanides' account of the Great Inflation, a rule fed a mismeasured real-time natural rate producing a sustained bias, though he locates the mismeasurement in $\phi_x x_t$ where the mechanism here is in the intercept; Orphanides & Williams on natural-rate misperception under learning is the closer antecedent.

4.3 The amplification result, and what it does and does not identify

The nominal endpoint the market prices is $r^{*m} + \pi^* + \delta/(\phi_\pi - 1)$; the endpoint the FOMC publishes is $r^{*F} + \pi^*$. The difference is

$$\boxed{\underbrace{i_t^{*m} - i_t^{*F}}_{\text{endpoint gap}} \;=\; \frac{\phi_\pi}{\phi_\pi - 1}\,\delta_t} \tag{10}$$

Identification. Given the published i^{*F}, the market's expected terminal rate reveals r^{*m} — but only up to the perceived ϕ_π. Equation (10) is one equation in two unknowns: the same endpoint gap is consistent with a small δ and a weakly-responding perceived Fed, or a large δ and an aggressive one. Separating them requires observing δ and the endpoint gap independently, which current data do not allow: the SPD's longer-run PCE median is 2.0 in all 91 surveys, so the survey gap measures δ itself and the amplified gap does not appear in it, and it cannot be taken from prices either, since far-forward risk-neutral rates load heavily on the current short rate. The result is best read as an identification statement — the published dot and the market's terminal rate differ by a *multiple* of the disagreement, and the multiple is the perceived Taylor coefficient — independent of the calibration below.

The magnitude claim. A small disagreement about r^* implies a large disagreement about where nominal rates would settle, with the multiplier set by the perceived ϕ_π:

ϕ_π	inflation bias per 1 pp of δ	endpoint gap per 1 pp of δ	endpoint gap when sd(δ) = 0.105
1.25	4.00	5.00	52 bp
1.50	2.00	3.00	32 bp
2.00	1.00	2.00	21 bp
2.50	0.67	1.67	18 bp

These are steady-state numbers, attained only if the market expects the disagreement never to be resolved — and that steady state is self-defeating. Reaching it requires the Fed to observe inflation running $\delta/(\phi_\pi - 1)$ above target indefinitely — at $\phi_\pi = 1.5$, twice the size of δ — and never to revise r^{*F}.

Key references. Bauer, Pflueger & Sunderam (2024), *QJE*. Bauer & Rudebusch (2020), *AER*. Caballero & Simsek (2022), *AER*. Cao, Crump, Eusepi & Moench, FRBNY SR 934. Cho & Williams (2026), Liberty Street Economics. Cieslak, McMahon & Pang, SSRN 4920858. Engstrom, FEDS 2026-026. Gürkaynak, Sack & Swanson (2005). Gürkaynak, Sack & Wright (2007; 2010), *JME*; *AEJ:Macro*. Hanson, Lucca & Wright (2021), *QJE*. Hartley (2025), Mercatus WP. Hillenbrand (2025), *RFS*. Holston, Laubach & Williams (2017), *JIE*; (2023), FRBNY SR 1063. Joslin, Singleton & Zhu (2011), *RFS*. Kozicki & Tinsley (2001), *JME*. Lou, Pintér, Üslü & Walker, BIS WP 1232. Lucca & Moench (2015), *JF*. Orphanides (2003), *JME*. Orphanides & Williams (2002), *BPEA*. Pan & Peng, SSRN 4764451. Swanson (2021), *JME*.